

Chapter 163

Resolve FX Revival

This category consists of plugins that let you fix common technical, damage, and quality problems that bedevil programs being finished, remastered, or restored.

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Automatic Dirt Removal (Studio Version Only)

The Automatic Dirt Removal plugin uses optical flow technology to target and repair temporally unstable bits of dust, dirt, hair, tape hits, and other unwanted artifacts that last for one or two frames and then disappear. All repairs are made while maintaining structurally consistent detail in the underlying frame, resulting in a high quality restoration of the image. Fortunately, despite its sophistication, this is a relatively easy plugin to use; just drop the plugin on a shot, adjust the parameters for the best results, and watch it go.



(Top) Original image, (Bottom) Using Automatic Dirt Removal

NOTE: This plugin is less successful with vertical scratches that remain in the same position for multiple frames and is completely ineffective for dirt on the lens that remains for the entire shot.

Main Controls

The primary controls used to adjust how much dirt is removed from the image.

- **Motion Estimation Type:** Lets you choose from among None, Faster, Normal, and Better. This tunes the tradeoff between performance and quality.
- **Neighbor Frames:** Lets you choose how many frames to compare when detecting dirt. Choosing more frames of comparison takes longer to process, but usually results in finding more dirt and artifacts.
- **Repair Strength:** This slider lets you choose how aggressively to repair dirt and artifacts that are found. Lower settings may let small bits through that may or may not be actual dirt, while higher settings eliminate everything that's found.
- **Dirt Size Threshold:** This slider lets you tune how large a detected bit of dirt must be to be removed. Raising this parameter lets you omit things like film grain from the operation but may allow smaller bits of dirt through.
- **Show Repair Mask:** This checkbox lets you see the dirt and artifacts that are detected by themselves, so you can see the effectiveness of the results as you fine tune this filter.

Fine Controls

These controls let you fine tune the effect in an effort to perfect the tradeoff between removing dirt successfully and preserving true image detail.

- **Motion Threshold:** This slider lets you choose the threshold at which pixels in motion are considered to be dirt and artifacts. At lower values more dirt may escape correction, but you'll experience fewer motion artifacts. At higher values, more dirt will be eliminated, but you may experience more motion artifacts in footage with camera or subject motion.
- **Edge Ignore:** This slider lets you exclude hard edges in the picture from being affected by dirt or artifacts that are removed. Higher values omit more edges from being affected.

Chromatic Aberration Removal (Studio Version Only)

A Revival category plugin that lets you manually correct the slight color fringing that results from chromatic aberration in a lens. Show Estimated Fringes checkboxes for the RGB channels display an “alignment guide” that visually isolates each of the types of fringing against gray.





(Previous page) The original image in close up, showing chromatic aberration,
(Above) The image with Estimated Red Fringes enabled, letting you see the specific problems as differences between red and cyan stripes

Advanced Options

Advanced Options provide additional parameters for problem shots.

- **Lens Center:** Center X and Y parameters let you offset the center of the lens if you're dealing with a reframed and re-rendered shot. You can also drag the lens center on-screen control in the Viewer to reposition.
- **Stronger Correction:** Shows the location of image features that resemble fringing.

Estimation Options

The Estimation options are used solely to highlight and identify areas of fringing. They do not affect the final output of the image. They only become available when one of the Show Estimated Fringes boxes is checked in the Aberration Correction section.

- **R/C, G/P, B/Y Balance:** These tools adjust the incoming balance between their respective colors to better identify hard to see fringing.
- **Brightness:** Magnifies the fringe indicators that are displayed when you turn on either of the Estimated Fringes checkboxes.

Aberration Correction

These tools let you make manual adjustments to correct aberration issues.

- **R/C, G/P, B/Y Scale:** Adjust these sliders to eliminate the fringing from their respective colors.
- **R/C, G/P, B/Y Edge:** Adjust to compensate for the difference in fringing due to the curvature of the lens' edges.
- **Show Estimated Fringes:** Checking this box will show just the estimated fringes over a gray background. It will also activate the Estimation Options sliders, which will let you further highlight and identify areas of fringing.

This makes it easy to make manual adjustments to correct the problem, using the Scale and Edge controls to individually adjust Red/Cyan, Green/Purple, and Blue/Yellow fringing.



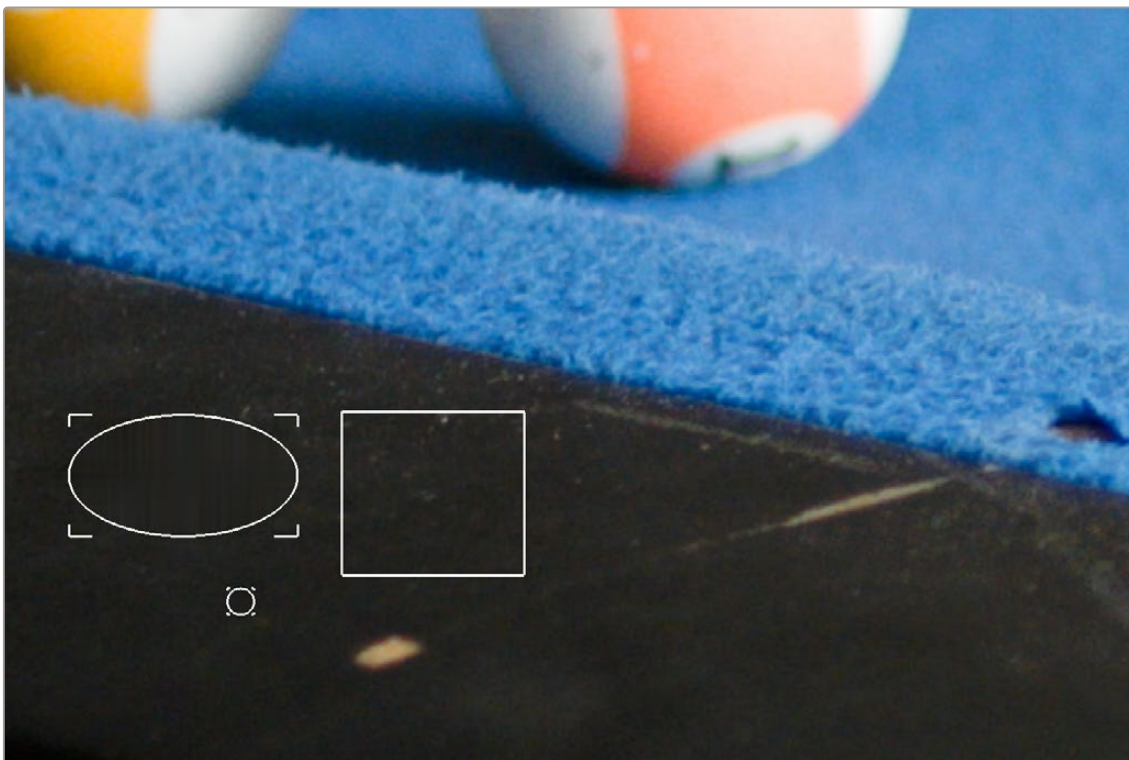
(Top) The original image in close up, showing chromatic aberration as a cyan fringe along the right of the smokestack and as a red fringe along the closest corner of the building, (Bottom) The corrected image in close up; this fringe is gone.

Dead Pixel Fixer (Studio Version Only)

If you have clips that were shot on a camera with one or more “dead” or “stuck” pixels in the sensor, you may have black or white spots that are fixed in place in the image. This filter is designed to let you place patches on each dead or stuck pixel, identifying them so you can use different methods of fixing the problem.

In many respects, the Dead Pixel Fixer is similar to the Dust Buster, however the Dust Buster effect is designed to repair transient bits of dust and dirt that only last for a frame or two, whereas the Dead Pixel Fixer is designed to work on blemishes that are fixed in place for the duration of a clip.

To fix dead or stuck pixels, apply the Dead Pixel Fixer filter, make sure the OFX onscreen controls are enabled in the viewer, and then click on each pixel you need to fix with the mouse to place a patch on it. You can click anywhere on the image to place as many patches as you like, there's no limit. You can also Option-click to delete patches you no longer need. To move any patches, simply drag it to another location.



Multiple dead pixel removal patches

When you place multiple patches, you can click to select whichever patches you want to adjust the controls for. Each patch can have different control settings.

General

These top level controls let you choose how to draw patches with which to repair blemishes in the frame, whether or not to show their on-screen controls, and how.

- **Draw Rect:** Click and drag to place a rectangular patch of any size from one corner to the opposite corner. Once drawn, you can click on any edge of the overlay and drag to reposition it.
- **Draw Ellipse:** Click and drag to place an elliptical patch of any size from one corner to the opposite corner. Once drawn, you can click on any edge of the overlay and drag to reposition it.
- **Place Patch:** Lets you place small patches specifically for tiny details such as dead pixels. When you choose this option, a New Patch Size slider appears that lets you adjust the size of the patches you're about to place prior to placing them. Once drawn, you can click on the edge of any patch and drag to reposition it.
- **New Patch Size:** (only when Place Patch is selected) Lets you choose how large of a patch to place using the Place Patch tool.
- **Show Patches:** This checkbox lets you show or hide the onscreen outline of every patch in the Viewer.
- **Hide During Interaction :** Hides the onscreen outline of every patch in the Viewer while you're moving a patch; this makes it easier to see the effect of moving the patch on the image, without the outline getting in the way.

Patch Type

There are three methods you can use to attempt to fix the contents of a selected patch. When drawing new patches, the currently selected Patch Type will determine what the next patch will be. When you've selected an existing patch, changing the Patch Type will change how that patch works.

- **Spatial:** Automatically fills the interior of the selected patch with pixels drawn from the surroundings of the patch, using the Fill Method. This works well for small blemishes, but for large blemishes a pattern might be discernible, which gives away the effect.
- **Clone:** Clone mode copies part of the image to fill a shape or patch placed over the thing you want to remove. In this mode, clicking and dragging to place a Rect or Ellipse over an imperfection is followed by a second click to place the sample region you want to clone. Clicking once to place a Patch will be followed by a second click to place the sample region. Selecting an existing shape or patch and choosing Clone lets you click on the shape to position the clone region. The sample region is indicated by a dotted shape that's connected to the original shape.
- **Blend Clone:** Operates similarly to Clone mode, except the copied part of the image that fills the Rect, Ellipse, or Patch is blended with the image to integrate it more softly.
- **Fill Method:** When using the Spatial Patch Type, the fill method determines how the blemish in the image is repaired.

Grid: Samples the pixels surrounding the Rect, Ellipse, or Patch, and blurs them inward both horizontally and vertically. Extremely effective for tiny blemishes. For larger blemishes, a grid-like pattern may emerge.

Horizontal: Samples the pixels to the left and right of the Rect, Ellipse, or Patch, and blurs them inward.

Vertical: Samples the pixels to the top and bottom of the Rect, Ellipse, or Patch, and blurs them inward.

Patchy: Samples pixels from all around the Rect, Ellipse, or Patch, and expands and blurs them together to create a soft, non-uniform region with which to repair the blemish. Doesn't have the patterning of the grid methods of fill, but produces an extremely smooth result.

Smooth: Simply uses a gaussian blur to repair the blemish.

Patch Options

These options let you customize the effect of a patch filling over a blemish.

- **Variability:** (Spatial patches only) Raising this parameter lets you make the Fill Method less uniform.
- **Soft Edges:** Lets you soften the edges of the patch.
- **Size Adjust:** Lets you change the size of a patch after its creation.
- **Mute:** Lets you turn a particular patch on or off via keyframing. Useful for blemishes that only show up for a few pixels of a shot.
- **Delete Patch:** Deletes the currently selected patch; you can also Option-Click any patch to delete it as well.

Advanced Controls

These controls let you customize the UI of this effect.

- **Clone Sticks to Mouse:** When enabled, drawing or placing a Rect, Ellipse, or Patch in Clone or Blend Clone mode immediately switches to positioning the sample region, making it faster to use.
- **New Patches Stay Selected:** When enabled, patches stay selected after you draw them, in preparation for further customization.
- **UI Line Thickness:** Lets you choose how thick the on-screen outlines should be.
- **Patches Stick to:** This parameter lets you deal with fixing dead pixels or blemishes on clips that have been stabilized.

Source: Patches you place stay put, unless you apply motion stabilization to the shot, in which case the patches are transformed along with the image so they remain “stuck” to the feature in the source clip they’re fixing.

Scene: If you’re eliminating a blemish on a moving subject, you can use the FX Tracker to track the thing you’re fixing, so the patch follows along with it.

- **Output Mode:** This lets you see different representations of the patched effect you’re creating. There are four options.

Patched Result: The final result, with each patch repairing the blemish it’s covering.

Patch Locations: Shows a key where each placed patch is white against black.

Differences: Shows the difference of each patch against the original image.

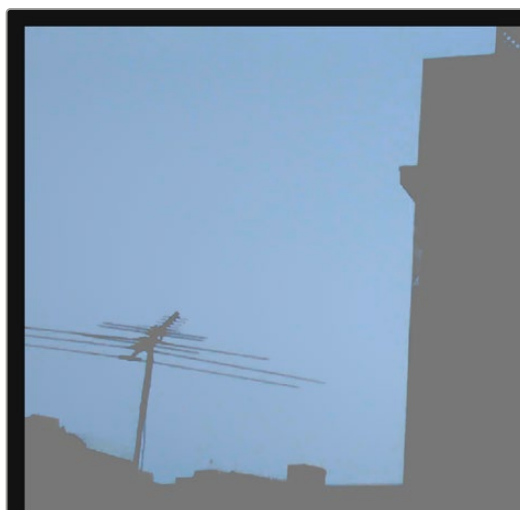
Difference Magnitude: Shows a more pronounced Differences representation.

Deband (Studio Version Only)

Low bit-depth media that has areas with shallow gradients of color, such as a sky or a wall, often exhibit color banding, seen as visible stripes, because there aren’t enough color values to smoothly represent the gradation from light blue to darker blue in the sky.



An example of banding in the sky of an image



Minimizing banding using the Deband filter

This filter is designed to isolate the edges of color banding and minimize them by dithering pixels from either side to soften the transition. This filter works best when applied to regions of the image that have been isolated with a secondary qualifier or window, otherwise you risk all edges within your image being dithered and thus softened (although this can be an interesting stylistic effect).

Deband Parameters

Deband controls include:

- **Edge Threshold:** Adjusts how well-defined an edge needs to be in order to be affected by this filter; lowering this excludes fainter edges, while raising this includes them.
- **Radius:** Lowers or raises the intensity of dithering in areas that will be affected.
- **Post Refine:** Narrows (by lowering) or widens (by raising) the areas affected by dithering.
- **Display Edge:** A checkbox that shows you a black and white high-contrast preview of which edges are being detected for dithering, which can help you fine-tune your results.

Deflicker (Studio Version Only)

The Deflicker plugin handles such diverse issues as flickering exposure in timelapse clips, flickering fluorescent lighting, flickering in archival film sources, and in certain subtle cases even the “rolling bars” found on video screens shot with cameras having mismatched shutter speeds. Two key aspects to this filter are that it only targets rapid, temporally unstable variations in lightness, and that it’s able to target only the areas of an image where flickering appears, leaving all other parts of the image untouched. As a result, this plugin can often repair problems once considered “unfixable.”



Original image with flicker bars in the blue screen



Result setting Deflicker to Fluoro Light, (clip courtesy Redline Films)

Main Parameters

By default, the top section of this plugin exposes a single control, which in many cases may be all you need.

- **Deflicker Settings pop-up menu:** The top two options, Timelapse and Fluoro Light, are presets that effectively eliminate two different categories of flickering artifacts. If neither of these presets is quite as effective as you’d hoped, a third option, Advanced Controls, opens up the Isolate Flicker controls at the heart of this plugin to let you tailor it further to your needs.

Temporal NR

Hidden by default, these controls only appear when you set Deflicker Setting to Advanced Controls, and let you choose how to detect motion in the scene so that flickering may be correctly addressed relative to the motion of subjects and items within the frame where it appears.

- **Frames Either Side:** Specifies the number of frames to analyze to determine what’s in motion. Higher values are not always better; the best setting is, again, scene dependent. The default is 3.
- **Mo.Est. Type:** Picks the method DaVinci Resolve uses to analyze the image to detect motion. Despite the names of the available options, which options will work best is highly scene

dependent. The default, Faster, is less processor intensive, but less accurate, however this can be an advantage and actually do a better job with high detail images that would confuse the Better option. Choosing Better is more accurate, but more processor intensive, and Better will try harder to match fine details which can sometimes cause problems. None lets you disable motion analysis altogether, which can work well (and will be considerably faster) in situations where there's no motion in the scene at all. The default is Better.

- **Motion Range:** Three settings, Small, Medium, and Large, let you choose the speed of the motion in the frame that should be detected.
- **Luma Threshold:** Determines the threshold above which changes in luma will not be considered flicker. The range is 0–100, 0 deflickers nothing, 100 applies deflickering to everything. The default is 100.
- **Chroma Threshold:** Determines the threshold above which changes in chroma will not be considered flicker. The range is 0–100, 0 deflickers nothing, 100 applies deflickering to everything. The default is 100.
- **Luma Chroma Same Threshold:** Lets you choose whether to gang the Luma and Chroma Threshold sliders or not.
- **Motion Threshold:** Defines the threshold above which motion will not be considered flicker.

Speed Optimization Options

Closed by default, opening this control group reveals two controls:

- **Reduced-Detail Motion checkbox:** On by default, reduces the amount of detail that's analyzed to detect flicker. In many cases, this setting makes no visible difference but increases processing speed. Disable this setting if your clip has fine detail that is being smoothed too aggressively.
- **Limit Analysis Area checkbox:** Turning this on reveals controls over a sample box that you can use to limit deflickering to a specific region of the image. This option is useful when:
 - a) Only one part of the image is flickering, so focusing on just that area speeds the operation considerably, or
 - b) Part of the image is being smoothed too much by deflickering that's fixing another part of the image very well.

Restore Original Detail After Deflicker

Closed by default, opening this control group reveals two controls:

- **Detail to Restore slider:** Lets you quickly isolate grain, fine detail, and sharp edges that should not be affected by the deflicker operation, preserving those fine details exactly.
- **Show Detail Restored checkbox:** Turning this checkbox on lets you see the edges that are detected and used by the Detail to Restore slider, to help you tune this operation.

Output

The Output pop-up menu lets you choose what Deflicker outputs, with options to help you troubleshoot problem clips. Here are the available options:

- **Deflickered Result:** The final, repaired result. This is the default setting.
- **Detected Flicker:** This option shows you a mask that highlights the parts of the image that are being detected as having flickering, to help you evaluate whether the correct parts of the image are being targeted. This mask can be very subtle, however.
- **Magnified Flicker:** This options shows you an exaggerated version of the Detected Flicker mask, to make it easier to see what the Deflicker plugin is doing.

Dust Buster (Studio Version Only)

This plugin is also designed to eliminate dust, dirt, and other imperfections and artifacts from clips, but it does so only with user guidance, for clips where the Automatic Dirt Removal plugin yields unsatisfactory results. This guidance consists of moving through the clip frame-by-frame and drawing boxes around imperfections you want to eliminate. Once you've drawn a box, the offending imperfection is auto-magically eliminated in the most seamless way possible. This works well for dirt and dust, but it also works for really big stains and blotches, as seen below.



(Previous page) Drawing a box around dirt in the original image, (Above) Result in the Dust Buster plugin

In many respects, the Dust Buster is similar to the Dead Pixel Fixer, however the Dust Buster effect is designed to repair transient bits of dust and dirt that only last for a frame or two, whereas the Dead Pixel Fixer is designed to work on blemishes that are fixed in place for the duration of a clip.

General

These top level controls let you choose how to draw patches with which to repair temporally unstable dust and dirt in the frame, whether or not to show their on-screen controls, and how.

- **Navigation Controls:** Four buttons let you jump to frames on which you've drawn patches. There are four buttons:
 - First Modified Frame:** Jumps the playhead to the first frame of that clip with a patch.
 - Previous Modified Frame:** Jumps the playhead to the last frame back with a patch.
 - Next Modified Frame:** Jumps the playhead to the next frame forward with a patch.
 - Last Modified Frame:** Jumps the playhead to the last frame of that clip with a patch.
- **Draw Rect:** Click and drag to place a rectangular patch of any size from one corner to the opposite corner. Once drawn, you can click on any edge of the overlay and drag to reposition it.
- **Draw Ellipse:** Click and drag to place an elliptical patch of any size from one corner to the opposite corner. Once drawn, you can click on any edge of the overlay and drag to reposition it.
- **Place Patch:** Lets you place small patches specifically for tiny details such as dead pixels. When you choose this option, a New Patch Size slider appears that lets you adjust the size of the patches you're about to place prior to placing them. Once drawn, you can click on the edge of any patch and drag to reposition it.
- **New Patch Size:** (only when Place Patch is selected) Lets you choose how large of a patch to place using the Place Patch tool.
- **Show Patches:** This checkbox lets you show or hide the onscreen outline of every patch in the Viewer.
- **Reset Frame:** Clears all patches from the current frame.

Patch Type

There are six methods you can use to attempt to fix the contents of a selected patch. When drawing new patches, the currently selected Patch Type will determine what the next patch will be. When you've selected an existing patch, changing the Patch Type will change how that patch works.

- **Auto:** The default method. Once you've drawn a bounding box, the two frames prior to and the two frames after the current clip will be analyzed and compared to the current image. The best of these 5 frames will be sampled to remove the imperfection in the current frame. Images two frames away are prioritized since that will avoid the appearance of frozen grain, but only if they're a suitable match to the content of the current frame.
- **+/- 1 Frame:** In this mode, if you draw a bounding box from left to right, the next frame will be drawn upon to remove the imperfection. If you draw a bounding box from right to left, the previous frame will be used.
- **+/- 2 Frames:** If you draw a bounding box from left to right, the image two frames forward will be drawn upon to remove the imperfection. If you draw a bounding box from right to left, the image two frames back will be used.
- **Spatial:** Automatically fills the interior of the selected patch with pixels drawn from the surroundings of the patch, using the Fill Method. This works well for small blemishes, but for large blemishes a pattern might be discernible, which gives away the effect.
- **Clone:** Clone mode copies part of the image to fill a shape or patch placed over the thing you want to remove. In this mode, clicking and dragging to place a Rect or Ellipse over an imperfection is followed by a second click to place the sample region you want to clone. Clicking once to place a Patch will be followed by a second click to place the sample region. Selecting an existing shape or patch and choosing Clone lets you click on the shape to position the clone region. The sample region is indicated by a dotted shape that's connected to the original shape.

- **Blend Clone:** Operates similarly to Clone mode, except the copied part of the image that fills the Rect, Ellipse, or Patch is blended with the image to integrate it more softly.
- **Fill Method:** When using the Spatial Patch Type, the fill method determines how the blemish in the image is repaired.

Grid: Samples the pixels surrounding the Rect, Ellipse, or Patch, and blurs them inward both horizontally and vertically. Extremely effective for tiny blemishes. For larger blemishes, a grid-like pattern may emerge.

Horizontal: Samples the pixels to the left and right of the Rect, Ellipse, or Patch, and blurs them inward.

Vertical: Samples the pixels to the top and bottom of the Rect, Ellipse, or Patch, and blurs them inward.

Patchy: Samples pixels from all around the Rect, Ellipse, or Patch, and expands and blurs them together to create a soft, non-uniform region with which to repair the blemish. Doesn't have the patterning of the grid methods of fill, but produces an extremely smooth result.

Smooth: Simply uses a gaussian blur to repair the blemish.

Patch Options

These options let you customize the effect of a patch filling over a blemish.

- **Variability:** (Spatial patches only) Raising this parameter lets you make the Fill Method less uniform.
- **Soft Edges:** Lets you soften the edges of the patch.
- **Size Adjust:** Lets you change the size of a patch after its creation.
- **Mute:** Lets you turn a particular patch on or off via keyframing. Useful for blemishes that only show up for a few pixels of a shot.
- **Delete Patch:** Deletes the currently selected patch; you can also Option-Click any patch to delete it as well.

Advanced Controls

These controls let you customize the UI of this effect.

- **Clone Sticks to Mouse:** When enabled, drawing or placing a Rect, Ellipse, or Patch in Clone or Blend Clone mode immediately switches to positioning the sample region, making it faster to use.
- **New Patches Stay Selected:** When enabled, patches stay selected after you draw them, in preparation for further customization.
- **UI Line Thickness:** Lets you choose how thick the on-screen outlines should be.
- **Output Mode:** This lets you see different representations of the patched effect you're creating. There are four options.

Patched Result: The final result, with each patch repairing the blemish it's covering.

Patch Locations: Shows a key where each placed patch is white against black.

Differences: Shows the difference of each patch against the original image.

Difference Magnitude: Shows a more pronounced Differences representation.

Frame Replacer (Studio Version Only)

The Frame Replacer allows you to reuse or blend adjacent frames together to help remove any corrupt frames or artifacts that show up for the duration of just a single frame. It's useful for fixing problems like a damaged film frame or a video frame that has pixelated digital breakup.

- **Replace This Frame:** Checking this box selects the specific frame in the clip that you want to replace. This will also automatically apply a keyframe at the appropriate location in the clip.
- **Replacement Method:** Allows you to choose the method by which the new frame is created.

From Previous: Replaces the selected frame with the one immediately before it, effectively duplicating the frame.

From Next: Replaces the selected frame with the one immediately after it, effectively duplicating the frame.

Blend Prev/Next: Replaces the selected frame with a blend of the frames immediately before and after it.

Optical Flow: Replaces the selected frame using DaVinci Resolve's optical flow technology with a blend of the frames before and after it.

Noise Reduction (Studio Version Only)

Based on the Noise Reduction controls in the Color page, the Resolve FX Noise Reduction is divided into two types of GPU-accelerated noise reduction designed to subdue noise in problematic clips. Both methods of noise reduction can be used separately or together, in varying amounts depending on the needs of the particular material you're working on.

Temporal NR Controls

The Temporal NR controls analyze images across multiple frames to isolate noise from detail. Motion estimation settings let you exclude moving subjects from this operation to prevent unwanted motion artifacts.

- **Frames Either Side:** The number of frames on either side of the current frame that you want averaged to separate detail from the noise. You can choose between 0 and 5 frames. 0 applies no frame averaging; higher values apply more frame averaging, at the expense of being significantly more computationally intensive. A higher frame setting may yield a better analysis, but may also yield unwanted artifacts if there are fast-moving images in the frame. A value of 1 may yield better results for fast-moving images. If you need to use higher frame values, but see artifacts, you can also try adjusting the Motion Threshold to fix the issue.
- **Motion Est. Type:** Picks the method DaVinci Resolve uses to detect motion in the image. The default, Faster, is less processor intensive, but less accurate. Choosing Better can effectively exclude motion more accurately, but is more processor intensive. None lets you disable motion estimation altogether, resulting in the application of Temporal NR to the entire image.
- **Motion Range:** Three settings, Small, Medium, and Large, let you set the speed of motion that Motion Estimation should expect to exclude. A Small setting assumes slow-moving subjects with little or no motion blur, allowing Temporal NR to affect more of the image at a given Motion Threshold setting. A Large setting assumes fast motion with blur occupying a larger area of the image, which excludes more of the image from Temporal NR at the same Motion Threshold setting. Choose the setting that gives you the best compromise between reducing noise and the introduction of motion artifacts when adjusting the Motion Threshold parameter.

Temporal Threshold Controls

The Temporal Threshold parameters allow you to control which image characteristics get more or less noise reduction.

- **Luma Threshold:** Lets you determine how much or how little Temporal NR to apply to the luma component of the image. The range is 0–100, where 0 applies no noise reduction at all, and 100 is the maximum amount. Too high a setting may eliminate fine detail from the image.
- **Chroma Threshold:** Determines how much Temporal NR is applied to the chroma component of the image. The range is 0–100, where 0 applies no noise reduction at all, and 100 is the maximum amount. Too high a setting may eliminate fine color detail from the image. However, you may find you can raise the Chroma Threshold higher than the Luma Threshold with less noticeable artifacting.
- **Luma Chroma Same Threshold:** Ordinarily, the Luma and Chroma Threshold parameters are ganged together so that adjusting one adjusts both. However, disabling this checkbox ungangs these parameters, allowing you to adjust different noise reduction amounts to each component of the image, depending on where the noise happens to be worst.
- **Motion:** Defines the threshold separating which moving pixels are in motion (above this threshold) versus which moving pixels are static (below this threshold). Using Motion Estimation, Temporal NR is not applied to regions of the image that fall above this threshold, to prevent motion artifacts by not applying frame-averaging to parts of the image that are in motion. Lower values omit more of the image from Temporal NR by considering more subtle movements. Higher values apply Temporal NR to more of the image by requiring faster motion for exclusion. You can choose between 0 and 100, where 0 applies Temporal NR to no pixels, and 100 applies Temporal NR to all pixels. The default value is 50, which is a suitable compromise for many clips. Be aware that if you set too high a Motion Threshold, you may see artifacts in moving parts of the image.
- **Blend:** Lets you dissolve between the image as it's being affected by the Temporal NR parameters (at 0.0) and the image with no noise reduction (100.0). This parameter lets you easily split the difference when using aggressive temporal noise reduction.

Spatial NR Controls

The Spatial NR controls let you smooth out regions of high-frequency noise throughout the image, while attempting to avoid softening by preserving detail. It's effective for reducing noise that Temporal NR can't.

- **Mode:** The Mode drop-down menu lets you switch Spatial NR between three different algorithms. All three modes of operation use the same controls, so you can switch between modes using the same settings to compare your results.
 - Faster:** Uses a computationally lightweight method of noise reduction that's good at lower settings, but may produce artifacts when applied at higher values.
 - Better:** Switches the Spatial NR controls to use a higher quality algorithm that produces greatly superior results to Faster, at the expense of being more processor intensive to render and not allowing you to decouple the Luma and Chroma Threshold sliders for individual adjustments to each color component.
 - Enhanced:** Does a significantly better job preserving image sharpness and detail when raising the Spatial Threshold sliders to eliminate noise. This improvement is particularly apparent when the Spatial Threshold sliders are raised to high values (what constitutes "high" varies with the image you're working on). At lower values, the improvement may be more subtle when compared to the "Better" mode, which is less processor intensive than the computationally expensive "Enhanced" setting. Additionally, "Enhanced" lets you decouple the Luma and Chroma threshold sliders so you can add different noise reduction amounts to each color component, as the image requires.

- **Radius:** Options include Large, Medium, and Small. A smaller radius offers greater real-time performance and can provide good quality when using low Luma and Chroma Threshold values. However, you may see more aliasing in regions of detail when using low NR Threshold values. Setting Radius to be progressively larger results in higher quality within areas of greater visual detail at high Luma and Chroma Threshold values, at the expense of slower performance. An NR Radius of Medium should provide suitable quality for most images when using medium NR Threshold settings. As with many operations, there's an adjustable tradeoff between quality and speed.

Spatial Threshold Controls

The Spatial Threshold parameters allow you to control which image characteristics get more or less noise reduction.

- **Luma:** Lets you determine how much or how little noise reduction to apply to the luma component of the image. The range is 0–100, where 0 applies no noise reduction at all, and 100 is the maximum amount. Too high a setting may eliminate fine detail from the image.
- **Chroma:** Lets you determine how much or how little noise reduction to apply to the chroma component of the image by smoothing out regions of high-frequency noise while attempting to preserve the sharpness of significant edge details. The range is 0–100, where 0 applies no noise reduction at all, and 100 is the maximum amount. Too high a setting may eliminate fine color detail from the image. However, you may find you can raise the Chroma Threshold higher than the Luma Threshold with less noticeable artifacting.
- **Luma Chroma Same Threshold:** Ordinarily, the Luma and Chroma Threshold parameters are ganged together so that adjusting one adjusts both. However, you can ungang these parameters to adjust different amounts of noise reduction to each component of the image. For example, if an image softens too much at a certain level of noise reduction, but you find more color speckling than luma noise, you can lower the Luma Threshold to preserve detail while raising the Chroma Threshold to eliminate color noise.
- **Blend:** Lets you dissolve between the image as it's being affected by the Spatial NR parameters (at 0.0) and the image with no noise reduction (100.0). This parameter lets you easily split the difference when using aggressive spatial noise reduction.

Global Blend

- **Blend:** Lets you dissolve between the image with no noise reduction (1.0) and the image with both Spatial NR and Temporal NR at their current settings (0.0).

Using Noise Reduction

The following procedure suggests a method of using the Noise Reduction (NR) parameters to achieve a controlled result.

Applying noise reduction to an image:

- 1 Enable Temporal NR by choosing 1 to 5 frames from the Number of Frames pop-up menu. Keep in mind that more frames dramatically increase the render time of this effect, while it may or may not significantly improve the result, depending on your material.
- 2 Choose options from the Motion Est. Type and Motion Range drop-down menus corresponding to how much motion is in the image. If there's a lot of motion, you may need to choose Better and Large. If there's not very much motion, lesser settings may suffice.

- 3 With Luma and Chroma Threshold linked, slowly raise either parameter until you just start to see a reduction in noise within nonmoving areas. Then make smaller adjustments to determine the maximum amount you can add without creating artifacts or overly softening detail.
- 4 If there's obviously more chroma than luma noise in the image, you can disable Luma/Chroma linking at a satisfactory level of luma noise reduction, and then raise the Chroma Threshold to address color speckling in the picture.
- 5 Suppose you're not satisfied with the tradeoff between the maximum possible threshold of noise reduction and the prevention of motion artifacts. In that case, you may want to adjust the Motion Threshold setting, lowering it to omit more of the motion from the noise reduction operation, or raising it to include more motion. If you're still not satisfied, you can also try better Motion Est. Type and Motion Range settings.

Keep in mind that the strength of Temporal NR is to reduce noise in unmoving parts of the image. When you've achieved the best tradeoff between noise reduction in the still areas and avoidance of motion artifacts in the moving areas of the image, it's time to turn to Spatial NR to further eliminate noise throughout the rest of the picture.

- 6 Enable Spatial NR by raising either the Luma or Chroma Threshold parameters, which are linked by default, until you strike a suitable balance between the reduction of noise and an unwanted increase in image softness.
- 7 It's recommended to choose the Enhanced option from the Spatial NR mode pop-up, as it will yield the best possible results. However, this can be processor-intensive, so if you need better real-time performance, you can switch the mode to Faster and compare results.
- 8 If there's obviously more chroma than luma noise in the image, you can disable Luma/Chroma linking at a satisfactory level of luma noise reduction, and then raise the Chroma Threshold to apply more aggressive Spatial NR to address color speckling in the picture.
- 9 If you've had to use a high Spatial NR Luma or Chroma Threshold setting to reduce noise visibly, and areas of detail look a bit chunky or aliased, you can choose a larger setting from the Radius pop-up menu to enable a more detailed analysis of the scene.

This will result in higher visual quality, but larger NR Radius settings are more processor-intensive and may reduce real-time performance if you don't have adequate GPU resources available to your system.

- 10 If you've found suitable noise reduction settings, but the result is too aggressive and makes the image appear too processed, you can try raising the Spatial NR and/or Temporal NR Blend parameters to fade between the noise reduction added by each set of controls, and the image as it was before you added noise reduction.

Try Applying Temporal NR First, then Applying Spatial NR

Because Temporal NR analyzes multiple frames for its noise isolation, it tends to be better at preserving detail accurately in regions of the image where there's little motion. If you try applying Temporal NR first and get a successful result, even if only in part of the image, you may reduce how much Spatial NR you have to apply, thus improving the overall quality of your final result.

Keep in mind that while Temporal NR does a great job in unmoving parts of an image but is less effective when dealing with subjects in motion, Spatial NR can reduce noise everywhere in the frame falling below its threshold, even when there's motion. Ultimately, a combination of the two is almost always going to be a winning combination.

Spatial NR Radius, How Large Should You Go?

Larger NR Radius settings can dramatically improve the quality of high-detail regions in shots where you're using aggressive Spatial noise reduction, but it's not necessary to always jump to the large Radius setting, which provides the highest precision. In many cases, when evaluating an image that you're applying noise reduction to, you may not be able to perceive the additional quality. You'll waste processing time on an unnecessary level of correction.

It's a good idea to evaluate the full-frame image on a large enough display to see the noise you're working on within the viewing context of the intended audience. Zooming really far into a clip while applying noise reduction may encourage you to use higher quality settings than are necessary because an excessively enlarged detail of an image lets you see subtle changes that you wouldn't notice at actual size.

Object Removal (Studio Version Only)

A Revival category plugin that's best used in the Color page, Object Removal uses the DaVinci Neural Engine to attempt to remove an object in the frame as automatically as possible. This plugin works best when removing a moving object that passes over a temporally stable background, or dirt on the lens of a shot where the camera is in motion. Smaller objects get better results than larger objects, but your results really depend on the footage. Here's a simple procedure that shows how to do this.

General

Show Mask Overlay: Lets you see the actual mask used, highlighted in red, for the Object Removal.

Analysis

These controls are used to analyze the motion of the scene.

- **Scene Analysis:** Press this button to analyze the clip using the selected parameters and replace the object with the appropriate background. You will need to press this button again if you change any of the parameters in this section.
- **Assume No Motion:** If the camera has been locked off for the shot while the object moves through it, check this box to greatly simplify the background analysis.
- **Scene Mode:** Provides different methods of analyzing the scene, for improving the analysis of how the area that needs to be replaced moves, to best determine how to fill the hole left by the object being removed.

Background: Analyzes the entire image except for the object region. Boundary analyzes the boundary area surrounding the object region. Object is for analyzing an object that moves with the background, like a sticker that's on a window while the camera moves.

Boundary: Analyzes the boundary area surrounding the object region.

Object: For analyzing an object that moves with the background, like a sticker that's on a window while the camera moves.

- **Analysis Boundary:** Sets the object's boundary size for analysis.
- **Show Scene Mask Overlay:** Toggles the scene mask overlay on or off, letting you see what's behind the mask.

Render

These parameters let you adjust how the Object Removal is blended into the background. The Object Removal plugin is highly footage dependent, and you won't always get a good result this easily. Problems with the result are shown via gray, either gray fringing or solid gray filling the replacement window. Gray shows you where the current settings are failing to find background content with which to fill in the patch you're removing.

- **Search Range:** If you notice while playing through the analyzed result that the object removal mask has gray fringing on some frames, you can try adjusting the "Search Range" slider, which is the distance, in frames, from the current frame that the Object Removal plugin is searching for replacement image detail.
For example, if the Search Range is 20, it searches +/-20 frames from the current location, or 40 frames total. The allowance of 10 frames means we look at every fourth frame. You will generally get the best results for the smallest range that gives an acceptable result.
- **Blend Mode:** If the patch is successfully filled, but the result isn't blending well with the background, you can try changing the Blend mode. The default is Linear, which is a simple cloning operation, but you can also choose Adaptive Blend, which can provide better results except in certain situations where the edges of the replacement patch have a different color or brightness than the background.

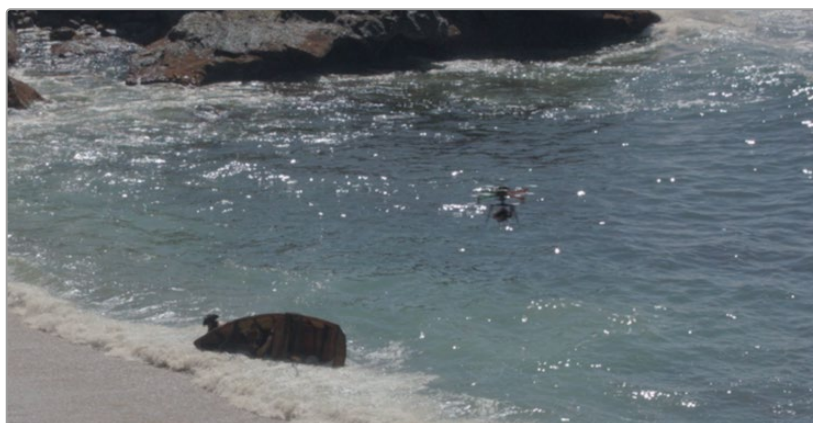
Clean Plate

If you're noticing that the object removal mask is filled entirely with gray on some frames, this means that background fill couldn't be easily generated for those frames. In this case, you can try clicking the "Build Clean Plate" button.

- **Clean Plate Source:** Lets you set the source of the clean plate. Options are Gray Image, which is effectively no source; Internal, which takes a "best guess" approach to generating a background with which to fill the frame and integrates this with frames that could be successfully filled in; and External, which lets you connect your own plate from another node.
- **Build Clean Plate:** Executes the building of a new clean plate background based on the Clean Plate Source parameter set.
- **Show Clean Plate:** Checking this box lets you see behind the mask to the clean plate behind it.

To remove a moving object from a clip:

- 1 In this example, a drone is flying through a long shot that's being simultaneously recorded. We'll remove the drone using a window to identify the feature to be removed using the Object Removal plugin.



The original shot with a drone that needs to be removed

In simple cases, it's often easiest to apply the Object Removal effect to a Corrector node, so you can use a window or qualifier within that node to isolate the feature you want to remove. That's what we'll do in this example.

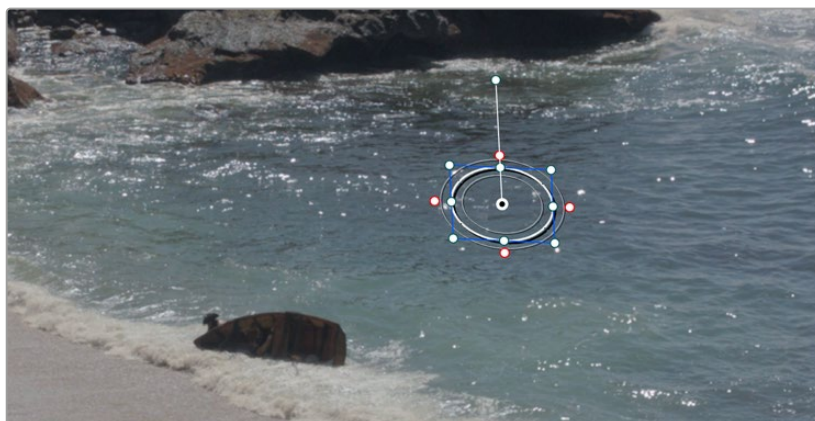
- 2 Use the Window palette to draw a window around the object that needs to be removed. You'll get the best results using windows or masks that hug the feature being removed fairly closely.
- 3 Track or keyframe the window to move with the feature you're removing. Again, you'll get better results the closer your window hugs the object being removed, and it's good to have some softness at the edge of this window.



The object that needs to be removed is isolated with a window

- 4 Drag and drop the Object Removal plugin onto the node in which you've just isolated the feature to be removed.
- 5 Click the Scene Analysis button, and wait for the analysis to finish. If the object you're removing is moving but the camera is locked, you can turn on the "Assume No Motion" checkbox to improve your results in this case.

If your footage is ideal for object removal, the object will disappear once analysis is complete, replaced by a seamless background derived from detail found on neighboring frames.



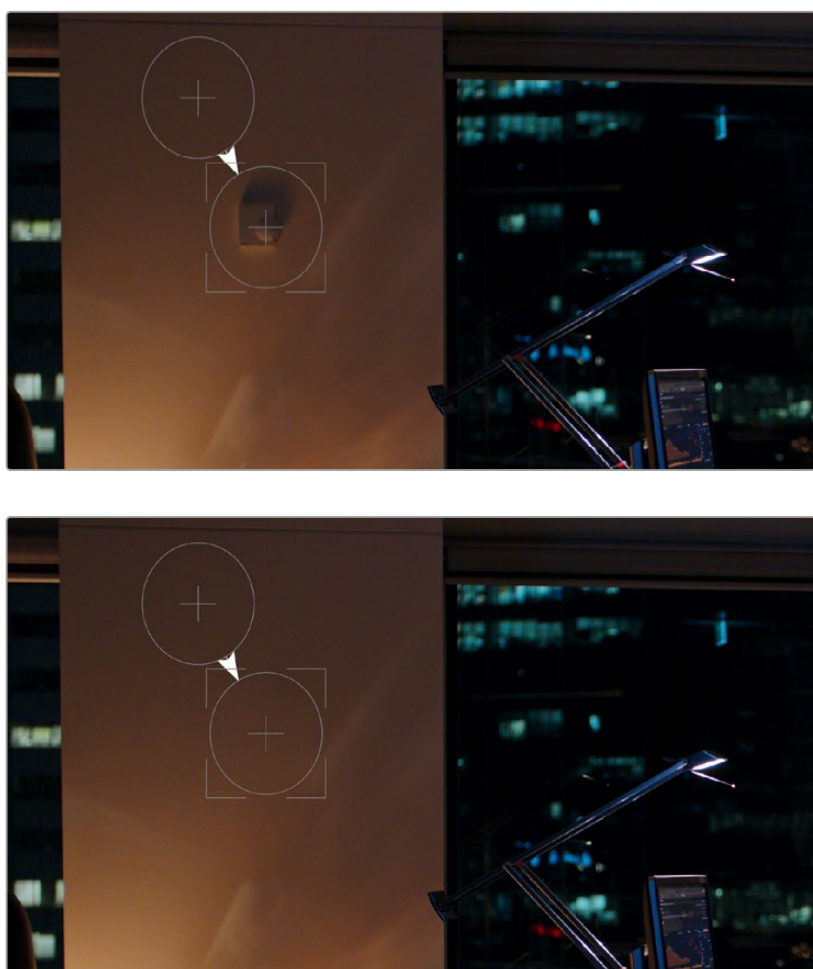
The result after object removal analysis is complete

Patch Replacer (Studio Version Only)

The Patch Replacer is a quick fix when you need to “paint out” an unwanted feature from the image. For those of you who’ve been using windows and Node Sizing to do small digital paint jobs, this plugin offers more options and a streamlined workflow.

On adding the plugin, an onscreen control consisting of two patches appears, with an arrow connecting them indicating which patch is being copied into the other. The patch behind the arrow is the “source” patch, used to sample part of the image, and the patch in front of the arrow is the “target” patch, used to cover up the unwanted feature using pixels from the source patch.

To use the Patch Replacer, simply drag the target patch over the feature you want to obscure, resize it to fit using the corner controls (the source patch is automatically resized to match), and then drag the source patch to an area of the image that can convincingly be used to fill the target patch.



(Top) Original image, (Bottom) Removing the thermostat with the Patch Replacer

The source and target patches can be motion tracked using the FX tracker, so this tool is effective even if the subject or camera is moving.

Main Controls

The Fill-in Method pop-up menu is arguably the most important, as it defines what method to use to fill the destination patch with whatever is in the source patch. The rest of the primary controls work differently depending on which Fill-in method you choose.

- **Clone:** Simply copies the source patch into the target patch. When Clone is selected, the Replacement Detail slider (which defaults to 1) lets you fade out the source patch, while Region Shape lets you choose a different kind of shape to use, and Blur Shape Edges lets you feather the edge of this operation, to more convincingly blend the source with the target area.
- **Blend Clone:** A much more sophisticated method of obscuring the target area using pixels from the source patch, and in many cases will yield better results more quickly than cloning. The source patch is copied into the target patch in such a way as to combine the source detail with the lighting found inside of the target area, creating in most instances a fast, seamless match. The Keep Original Detail checkbox, when turned on, merges detail from the source and target patches to create a composite, rather than a fill. The Variability slider lets you adjust how solid the center of the patch is. The Blur Shape Edges slider works a bit differently with Blend Clone selected, but the idea is the same, feathering the effect from the outside in to obscure instances where there's a noticeable border around the target area.
- **Fast Mask:** Eliminates the source patch, doing instead a quick neighboring pixel blend that works well with small patches but can betray a grid pattern on larger patches. Region Shape and Blur Shape Edges are both adjustable.
- **Blend Mask:** Eliminates the source patch, and essentially does the Blend Clone operation but without the source. Works well on small patches, but can become too blurred on larger patch sizes. Gives you access to the Variability slider of the Blend Clone.

Patch Positions

Source X and Y, Target X and Y, and Target Width and Height are provided as explicit controls both for numeric adjustment, should that be necessary, and also to allow for keyframing in case you need to change the position and/or size of the source and fill patches over time.

Keep in mind that the source and target patches can be motion tracked using the FX tracker, although two checkboxes, Source Follows Track and Target Follows Track, let you disable FX tracker match moving when necessary.

The Align Source to Target button moves the source patch directly over the target patch, allowing you to quickly reset and iterate the location of the source patch.

Onscreen Controls

The Control Visibility pop-up menu lets you choose whether the source and target onscreen controls are visible as you work. Show (the default) leaves all onscreen controls visible all the time. Auto Hide hides all onscreen controls whenever you're dragging one, letting you see the image as you adjust it without having these controls in the way. Hide makes all onscreen controls invisible, so you can see a clean version of the image with the effect, however you can still edit the effect if you remember where the controls are.